T1 According to the title, we can know:

$$A=\begin{pmatrix}-1 & 1 & 3\\ 12 & 4 & 10\end{pmatrix},\ b=\begin{pmatrix}20\\90\end{pmatrix},\ c^T=(-5,5,13)$$

$$B=\begin{pmatrix}1 & 0\\ 4 & 1\end{pmatrix},\ B^{-1}=\begin{pmatrix}1 & 0\\ -4 & 1\end{pmatrix},\ c_B^T=(5,0)$$

If we change the RHS of constraint 2 to $\bar{b}=70$

$$\therefore \bar{b}=\begin{pmatrix}20\\70\end{pmatrix} \qquad \therefore B^{-1}\bar{b}=\begin{pmatrix}1 & 0\\ -4 & 1\end{pmatrix}\begin{pmatrix}20\\70\end{pmatrix}=\begin{pmatrix}20\\-10\end{pmatrix}$$

$\because -10<0$ $\therefore$ the solution is NOT feasible.

Now use the dual simplex method, the dual problem is:

minimize $z=20y_1+70y_2$ $\qquad -z=-20y_1-20y_2$

s.t. $$\begin{pmatrix}-1 & 12\\ 1 & 4\\ 3 & 10\end{pmatrix}\begin{pmatrix}y_1\\y_2\end{pmatrix}\geq\begin{pmatrix}-5\\5\\13\end{pmatrix}$$

$y_1, y_2 \geq 0$.

Introduce the slack variable $y_3, y_4, y_5 \geq 0$

Iteration	BV	Row	Coefficient						RHS	Ratio
			z	y_1	y_2	y_3	y_4	y_5		
	z	(0)	-1	20	70	0	0	0	0	
0	y_3	(1)	0	1	-12	1	0	0	5	
	y_4	(2)	0	-1	-4	0	1	0	-5	
	y_5	(3)	0	-3	-10	0	0	1	-13.	
	z	(0)	-1	0	$\frac{10}{3}$	0	0	$\frac{20}{3}$	$-\frac{260}{3}$	
	y_3	(1)	0	0	$-\frac{46}{3}$	1	0	$\frac{1}{3}$	$\frac{2}{3}$	
1	y_4	(2)	0	0	$-\frac{2}{3}$	0	1	$-\frac{1}{3}$	$-\frac{2}{3}$	
	y_1	(3)	0	1	$\frac{10}{3}$	0	0	$-\frac{1}{3}$	$\frac{13}{3}$	

	z	(0)	-1	0	0	0	5	5	-90	
2	y_3	(1)	0	0	0	1	-23	8	16.	
	y_2	(2)	0	0	1	0	$-\frac{3}{2}$	$\frac{1}{2}$	1	
	y_1	(3)	0	1	0	0	5	-2	1	

$\therefore (1,1,16,0,0)$ is the optimal solution of dual problem.

$\therefore y^* = (1,1)$

$$\therefore \begin{cases} -x_1 + x_2 + 3x_3 = 20 \\ 12x_1 + 4x_2 + 10x_3 = 70 \\ x_1(-y_1 + 12y_2 + 5) = 0 \\ x_2(y_1 + 4y_2 - 5) = 0 \\ x_3(3y_1 + 10y_2 - 3) = 0 \end{cases} \Rightarrow \begin{cases} x_1 = 0 \\ x_2 = 5 \\ x_3 = 5 \end{cases}$$

$\therefore (0,5,5,0,0)$ is the optimal solution

T2 (a) According to the title, we can assume x_i is the number of skilled technicians in ith period, y_i is the number of new technicians in ith period.

So we can set up a LP like that:

$$\begin{aligned} \text{minimize} \quad & 2000(x_1+x_2+x_3+x_4+x_5). \\ \text{s.t.} \quad & 160x_1 - 50y_1 \geq 6000 \\ & 160x_2 - 50y_2 \geq 7000 \\ & 160x_3 - 50y_3 \geq 8000 \\ & 160x_4 - 50y_4 \geq 9500 \\ & 160x_5 - 50y_5 \geq 11000. \\ & x_t = x_{t-1} \cdot 0.95 + y_{t-1} \\ & x_1 = 50 \\ & x_1, x_2, \cdots, x_5, y_1, \cdots, y_5 \geq 0. \end{aligned}$$

(b) Please refer to the attachment for details.

(c) the primal problem's optimal solution is:

$\begin{cases} x_1 = 50 \\ y_1 = 0 \end{cases}$, $\begin{cases} x_2 = 47.5 \\ y_2 = 8.45 \end{cases}$, $\begin{cases} x_3 = 53.58 \\ y_3 = 11.45 \end{cases}$, $\begin{cases} x_4 = 62.35 \\ y_4 = 9.52 \end{cases}$, $\begin{cases} x_5 = 68.75 \\ y_5 = 0. \end{cases}$

the duel problem's optimal solution is:
(3900, 2000, 0, -481.93, -598.05, 0, 0, 9.64, 11.96, 16.24)

According to the answer of computer, we can know that the objective function values of primal and duel problem are equal to 564355.13, which means it is strong duality holds.

The dual variable corresponds to the "shadow prize" of the primal problem constriant. It represents the amount of cost increase for each additional unit of demand working hours in the case that the demand for working hours is met.

(d) Please refer to the attachment for details.